

Technical Cyber security Report

Analysis Using Metasploit and Nmap for Personal Local IP address

By

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REPORT OUTLINE

1. Executive Summary	3
2. Background and Objectives	3
3. Methodology	3
3.1 Tool Configuration	3
3.2 Execution Process	4
3.3 Monitoring and Analysis:	4
4. Findings and Analysis	5
4.1 Indicators of Compromise (IOCs):	5
4.2 Risk and Impact Assessment:	5
5. Recommendations	5
5.1 Immediate Remediation Actions:	5
5.2 Long-Term Mitigation:	5
6. Conclusion	5
Appendix: Additional Data	6

1. Executive Summary

- **Overview:**
 - The analysis with Metasploit using Nmap was conducted on my local IP address - 192.168.52.128 to discover vulnerabilities on my network. The tool provided number of open/closed ports, the status of the host system and the latency values of the network.
 - **Key Findings:**
 - Metasploit detected that the host was active and there were no open ports out of the 1000 ports scanned on that IP address. This shows that the IP address is properly secured.
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2. Background and Objectives

- **Project Context:**
 - The project aimed to assess the security posture of the personal IP address - 192.168.52.128 and identify the types of open port, the latency value of the network and the overall status of that IP address.
 - **Objective of the Tool Use:**
 - Metasploit using Nmap was utilised to scan the 1000 ports on the IP address 192.168.52.128 by running it in a Virtual machine (Kali Linux) to examine and assess the overall security posture of the network.
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3. Methodology

3.1 Tool Configuration

- **Metasploit using Nmap Configuration:**
 1. Open Kali Linux on your personal system
 2. Enter your credentials
 3. Open command prompt on the Kali Linux
 4. Enter ifconfig to get the laptop IP address

```
root@kali: /home/kali
File Actions Edit View Help
(kali@kali)-[~]
└─$ sudo su
[sudo] password for kali:
(kali@kali)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.52.128 netmask 255.255.255.0 broadcast 192.168.52.255
    inet6 fe80::742e:c932:1916:5746 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:92:fc:5b txqueuelen 1000 (Ethernet)
    RX packets 1619 bytes 118344 (115.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 39 bytes 6084 (5.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 86013 bytes 11856961 (11.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 86013 bytes 11856961 (11.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

5. Copy the IP address
6. Enter sudo su to run as root user and enter password
7. Search and open Metasploit in the kali linux
8. Enter kali linux password in shell
9. Wait until Metasploit installs on the VM
10. Enter the keyword db_nmap -sC -sV -vv 192.168.52.128 to scan the database of your IP address

```
-- ==[ metasploit v6.4.34-dev ]
-- ==[ 2461 exploits - 1267 auxiliary - 431 post ]
-- ==[ 1471 payloads - 49 encoders - 11 nops ]
-- ==[ 9 evasion ]

metasploit Documentation: https://docs.metasploit.com/

sf6 > db_nmap -sC -sV -vv 192.168.52.128
```

11. The result will be displayed.

3.2 Execution Process

- **File Execution in Sandbox:**

Upon execution, it was noticed that out of the 1000 tcp ports that was pinged, all of them had the ignored states which showed that they were all closed ports. The local host has a latency of 0.0000050s which is less than 1. This shows that the IP address is local and reachable.

3.3 Monitoring and Analysis:

- **System Monitoring:**

The scanning recorded that the 1 host was up. Which meant that the Operating system was fully functional.

- **Network Monitoring:**

The IP address was recorded that the 1 host was up with all its 1000 ports closed. The SYN stealth scan was run on 1000 total ports in 0.05s

4. Findings and Analysis

4.1 Indicators of Compromise (IOCs):

There was no IOC as all the 1000 tcp ports were closed.

4.2 Risk and Impact Assessment:

- **Potential Impact:**

If this IP address is exposed, there would be no major impact on the IP address as there is no open port on the IP address that can be exploited.

5. Recommendations

5.1 Immediate Remediation Actions:

- Ensure that there is monitoring of the IP address to checking for any form of open tcp ports and the latency value on that IP address.

5.2 Long-Term Mitigation:

- **Monitoring and Detection Enhancements:**

Recommend that there is continuous monitoring of the IP address to checking for any form of open tcp ports on that IP address in the future.

6. Conclusion

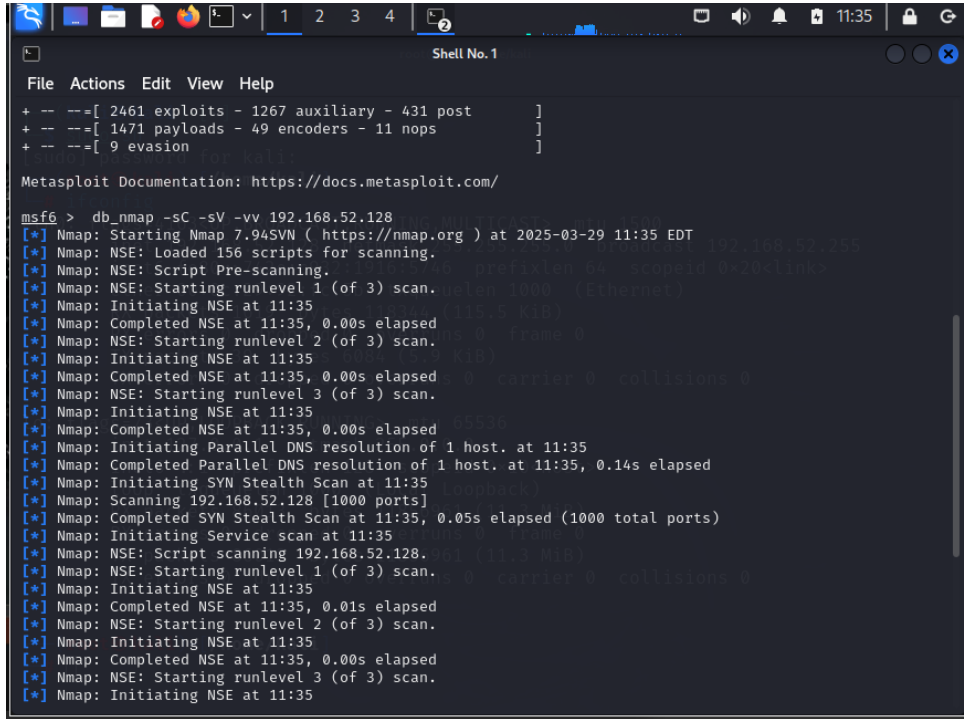
- **Summary of Findings:**

Metasploit using Nmap successfully identified that the local host IP address 192.168.52.128 is up because there was no open ports and had a latency value of less than 1.

Appendix: Additional Data

- Raw Logs and Tool Outputs:

Here is the raw result from Metasploit using nmap scanning

A screenshot of a Metasploit terminal window titled "Shell No. 1". The window shows the Metasploit interface with a menu bar (File, Actions, Edit, View, Help) and a status bar at the top displaying "1 2 3 4". The main content area shows the output of the command "db_nmap -sC -sV -vv 192.168.52.128". The output is a detailed log of the Nmap scan process, including script loading, NSE (Nmap Scripting Engine) execution, and the final scan results. The scan is performed on the IP address 192.168.52.128. The output includes the following information: Nmap version 7.94SVN, scan time 2025-03-29 11:35 EDT, NSE loaded 156 scripts, NSE script pre-scanning, NSE starting runlevel 1 (of 3) scan, NSE initiating NSE at 11:35, NSE completed NSE at 11:35, 0.00s elapsed, NSE starting runlevel 2 (of 3) scan, NSE initiating NSE at 11:35, NSE completed NSE at 11:35, 0.00s elapsed, NSE starting runlevel 3 (of 3) scan, NSE initiating NSE at 11:35, NSE completed NSE at 11:35, 0.00s elapsed, Nmap initiating Parallel DNS resolution of 1 host. at 11:35, Nmap completed Parallel DNS resolution of 1 host. at 11:35, 0.14s elapsed, Nmap initiating SYN Stealth Scan at 11:35, Nmap scanning 192.168.52.128 [1000 ports], Nmap completed SYN Stealth Scan at 11:35, 0.05s elapsed (1000 total ports), Nmap initiating Service scan at 11:35, Nmap NSE: Script scanning 192.168.52.128, Nmap NSE: Starting runlevel 1 (of 3) scan, Nmap NSE: Initiating NSE at 11:35, Nmap NSE: Completed NSE at 11:35, 0.01s elapsed, Nmap NSE: Starting runlevel 2 (of 3) scan, Nmap NSE: Initiating NSE at 11:35, Nmap NSE: Completed NSE at 11:35, 0.00s elapsed, Nmap NSE: Starting runlevel 3 (of 3) scan, Nmap NSE: Initiating NSE at 11:35.

```
msf6 > db_nmap -sC -sV -vv 192.168.52.128
[*] Nmap: Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-03-29 11:35 EDT 192.168.52.255
[*] Nmap: NSE: Loaded 156 scripts for scanning.
[*] Nmap: NSE: Script Pre-scanning.
[*] Nmap: NSE: Starting runlevel 1 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 2 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 3 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: Initiating Parallel DNS resolution of 1 host. at 11:35
[*] Nmap: Completed Parallel DNS resolution of 1 host. at 11:35, 0.14s elapsed
[*] Nmap: Initiating SYN Stealth Scan at 11:35
[*] Nmap: Scanning 192.168.52.128 [1000 ports]
[*] Nmap: Completed SYN Stealth Scan at 11:35, 0.05s elapsed (1000 total ports)
[*] Nmap: Initiating Service scan at 11:35
[*] Nmap: NSE: Script scanning 192.168.52.128.
[*] Nmap: NSE: Starting runlevel 1 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.01s elapsed
[*] Nmap: NSE: Starting runlevel 2 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 3 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
```

```
msf6 > nmap 192.168.52.128
[*] Nmap: Initiating SYN Stealth Scan at 11:35
[*] Nmap: Scanning 192.168.52.128 [1000 ports]
[*] Nmap: Completed SYN Stealth Scan at 11:35, 0.05s elapsed (1000 total ports)
[*] Nmap: Initiating Service scan at 11:35
[*] Nmap: NSE: Script scanning 192.168.52.128.
[*] Nmap: NSE: Starting runlevel 1 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.01s elapsed
[*] Nmap: NSE: Starting runlevel 2 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 3 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: Nmap scan report for 192.168.52.128
[*] Nmap: Host is up, received localhost-response (0.0000050s latency).
[*] Nmap: Scanned at 2025-03-29 11:35:01 EDT for 0s
[*] Nmap: All 1000 scanned ports on 192.168.52.128 are in ignored states.
[*] Nmap: Not shown: 1000 closed tcp ports (reset)
[*] Nmap: NSE: Script Post-scanning.
[*] Nmap: NSE: Starting runlevel 1 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 2 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 3 (of 3) scan.
[*] Nmap: Initiating NSE at 11:35
[*] Nmap: Completed NSE at 11:35, 0.00s elapsed
[*] Nmap: Read data files from: /usr/share/nmap
[*] Nmap: Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds
[*] Nmap: Raw packets sent: 1000 (44.000KB) | Rcvd: 2000 (84.000KB)
msf6 >
```